

Results

01 • VAR

several interrelated series

Table 1. Lag selection

Lag	AIC	BIC	HQ
1	6.67	6.88	6.75
2	6.64	6.98	6.77
3	6.57	7.05	6.76
4	5.98	6.60	6.23
5	5.96	6.71	6.25
6	5.87	6.76	6.22
7	5.55	6.58	5.96
8	4.94	6.10	5.40
9	4.65	5.95	5.16
10	4.69	6.13	5.25

Table 2. VAR coefficients (per equation)

	sales	visitors
sales.l1	-0.22	0.36
	(0.15)	(0.18)
	[-0.53, 0.08]	[-0.02, 0.73]
visitors.l1	0.46	0.08
	(0.12)	(0.15)
	[0.22, 0.70]	[-0.22, 0.37]
sales.l2	-0.22	0.15
	(0.14)	(0.18)
	[-0.52, 0.07]	[-0.21, 0.50]
visitors.l2	0.50	-0.03
	(0.13)	(0.16)
	[0.23, 0.77]	[-0.36, 0.30]
sales.l3	-0.02	0.32
	(0.15)	(0.18)
	[-0.32, 0.27]	[-0.04, 0.68]
visitors.l3	0.57	-0.16
	(0.14)	(0.18)
	[0.28, 0.86]	[-0.52, 0.20]
sales.l4	-0.38	0.14
	(0.14)	(0.17)
	[-0.66, -0.10]	[-0.21, 0.49]
visitors.l4	0.46	0.10
	(0.16)	(0.19)
	[0.15, 0.78]	[-0.29, 0.49]
sales.l5	-0.21	0.29
	(0.14)	(0.18)
	[-0.50, 0.08]	[-0.07, 0.64]
visitors.l5	0.24	-0.50
	(0.14)	(0.18)
	[-0.05, 0.53]	[-0.86, -0.15]
sales.l6	-0.41	0.06
	(0.14)	(0.17)
	[-0.70, -0.13]	[-0.29, 0.41]
visitors.l6	0.13	-0.53
	(0.17)	(0.20)
	[-0.20, 0.47]	[-0.94, -0.12]
sales.l7	-0.34	0.18
	(0.15)	(0.19)
	[-0.65, -0.04]	[-0.19, 0.55]
visitors.l7	-0.05	-0.42
	(0.17)	(0.21)
	[-0.39, 0.29]	[-0.84, -0.00]
sales.l8	0.10	0.30
	(0.16)	(0.20)

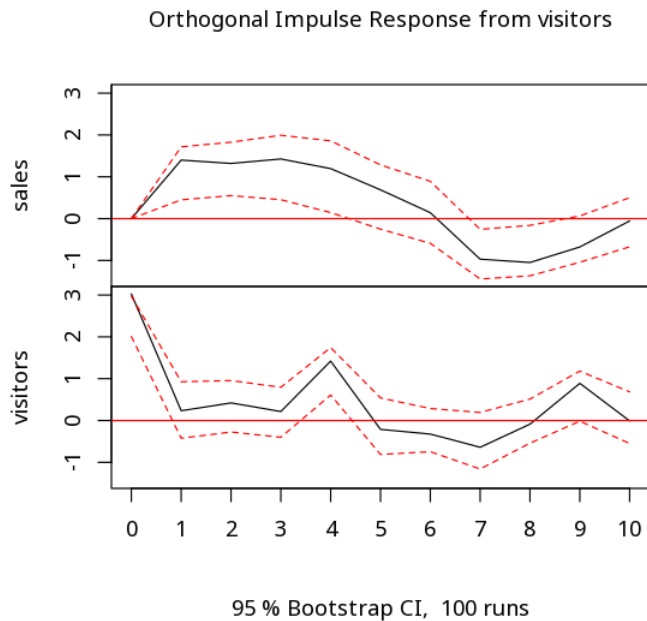
	sales	visitors
	[-0.22, 0.43]	[-0.10, 0.70]
visitors.l8	0.13 (0.16) [-0.19, 0.44]	-0.20 (0.19) [-0.59, 0.19]
sales.l9	0.36 (0.15) [0.06, 0.66]	0.24 (0.18) [-0.13, 0.61]
visitors.l9	0.18 (0.14) [-0.09, 0.45]	0.39 (0.17) [0.05, 0.73]
const	-261.13 (96.47) [-455.54, -66.71]	238.95 (118.72) [-0.32, 478.22]
Num.Obs.	63	63
R ²	1.00	.99
R ² Adj.	.99	.99
RMSE	2.18	2.68
Log.Lik.	-138.42	-151.50
Selected lag (p) = 9		
Max root modulus = 1.00 (stable)		
Portmanteau (serial) $\chi^2(28) = 112.35$, p < .001		

one column per equation (response series); each cell stacks the estimate, its (SE), and the [95% CI] — the per-coefficient t/p are dropped. orthogonalised (Cholesky) IRFs depend on the ordering of the series; a level VAR assumes stationarity. stability check: max companion-eigenvalue modulus < 1 indicates a stable VAR. serial-correlation check: the Portmanteau test (vars::serial.test, asymptotic) tests the null of no residual autocorrelation — a small p suggests remaining serial correlation (consider a higher lag order).

Table 3. Forecast-error variance decomposition

Variable	Impulse	Share
sales	sales	.50
sales	visitors	.50
visitors	sales	.26
visitors	visitors	.74

Figure. Dynamic response



How to read this test

A VAR models several series as functions of their joint past. The impulse-response plots show how a shock to one series propagates to the others over time; read each IRF together with its bootstrap confidence band, and note that orthogonalised (Cholesky) IRFs depend on the ordering of the series. Lag selection picks how many past periods to include. A level VAR assumes stationary series — difference them (or use a VECM) if they are not. The Portmanteau test (`vars::serial.test`) checks the residuals for remaining serial correlation: a small p suggests the lag order is too low to whiten the residuals. Method: R vars package (Pfaff, 2008; Lütkepohl, New Introduction to Multiple Time Series Analysis).

APA template: A VAR(9) model was selected by AIC; impulse-response functions are shown (Figure).

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